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Deep Learning and Its Applications (CO3133)

Semester 261 – Class CC01 – Group G-02

Assignment 3 – Milestone 2

M2 Draft Report

Knowledge-Based Visual Question Answering

Multimodal Deep Learning on A-OKVQA

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1 Task and data (summary of M1)

Task: multiple-choice visual question answering on A-OKVQA [1]. One sample is a COCO image [2], a question that needs commonsense or world knowledge, and 4 choices; the model picks one choice. Details of the data and the EDA are in the M1 proposal and `Assignment3/01_eda.ipynb`.

Item	Value
Dataset	`HuggingFaceM4/A-OKVQA`, revision `d1b0efa`; annotations Apache 2.0, COCO images
Labelled pairs	18,201 questions on 17,661 images (official test split has no labels)
Split (by image)	train 15,346 / validation 1,710 / test 1,145 (= official validation); seed 42
Modalities	image (modality B) and question + choices (modality A)
Missing modality	none (0 missing images or questions)
Proposal status	submitted; no formal decision returned yet, the group continued after checking the handbook rules

Table 1: Data and split used in M2.



Figure 1: Random training pairs; * marks the correct choice.

2 Methods of the draft

All models share one frozen encoder, CLIP ViT-B/32 [3]. Its outputs are computed once (`Assignment3/features.py`, 99 s for all images and 72 s for all texts on one A40) and cached: an image vector v, a question vector q and a vector c_j

for each choice, all 512-d and scaled to length 1. A small MLP (Linear $\rightarrow$ ReLU $\rightarrow$ Dropout 0.1 $\rightarrow$ Linear to 1 score) scores each choice; a softmax over the 4 scores and cross-entropy give the loss.

Model	**Input of the MLP for choice j**	**Trainable**
TextOnly (modality A)	$[q;\ c_j;\ q \odot c_j]$	787,457
ImageOnly (modality B)	$[v;\ c_j;\ v \odot c_j]$ (choices name the answers; no question)	787,457
LateFusion (simple fusion)	$[v;\ q;\ c_j;\ v \odot c_j;\ q \odot c_j]$	1,311,745

Table 2: The three M2 models. References without training: chance, the answer-text prior of the EDA, and zero-shot CLIP ($\cos(v,$ "question answer: choice")).

Training: Adam, learning rate 10^{-3}, batch 64, 20 epochs, checkpoint with the best validation accuracy, seeds 42–46, settings in `configs/a3.yaml`. The test split is used once per run. Metrics: multiple-choice accuracy (main) and the A-OKVQA direct-answer (DA) score of the chosen text, $\min(n/3, 1)$ with n the number of workers who gave it.

3 Preliminary results

Model	**Val. accuracy**	**Test accuracy**	**Test DA score**	**Test, shuffled images**
Chance		0.250		
Answer-text prior		0.312	0.281	
Zero-shot CLIP		0.536	0.438	
TextOnly	0.475 ± 0.003	0.433 ± 0.010	0.362 ± 0.008	0.433
ImageOnly	0.579 ± 0.005	0.572 ± 0.010	0.463 ± 0.009	0.322
LateFusion	$\mathbf{0.587} \pm 0.008$	$\mathbf{0.577} \pm 0.009$	$\mathbf{0.470} \pm 0.008$	0.343

Table 3: Results of the M2 models, mean $\pm$ SD over 5 seeds. "Shuffled images": every test image replaced by the image of another test question.

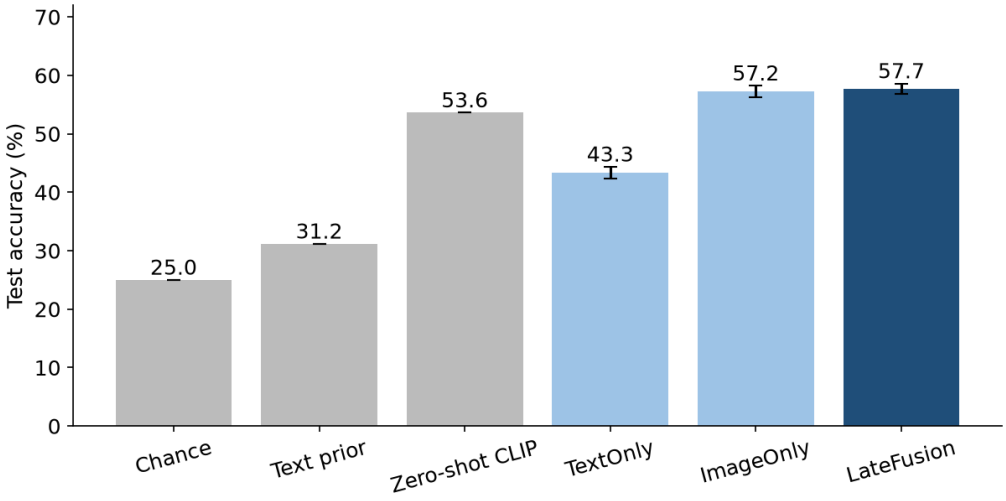


Figure 2: Test accuracy of the references and the M2 models (error bars: seed SD).

- **The image is the strong modality.** ImageOnly (57.2%) is far above TextOnly (43.3%); LateFusion is 14.4 points above TextOnly in all 5 seeds.

- **The question adds little on top of the image.** LateFusion is only 0.5 points above ImageOnly, less than the seed SD. With CLIP features, matching the image with the 4 choices already answers most questions.
- **The models use the image.** With shuffled images, accuracy falls by 23–25 points, below the text-only model.
- **Training helps a little over zero-shot.** Zero-shot CLIP already gets 53.6%; the trained late fusion adds about 4 points.

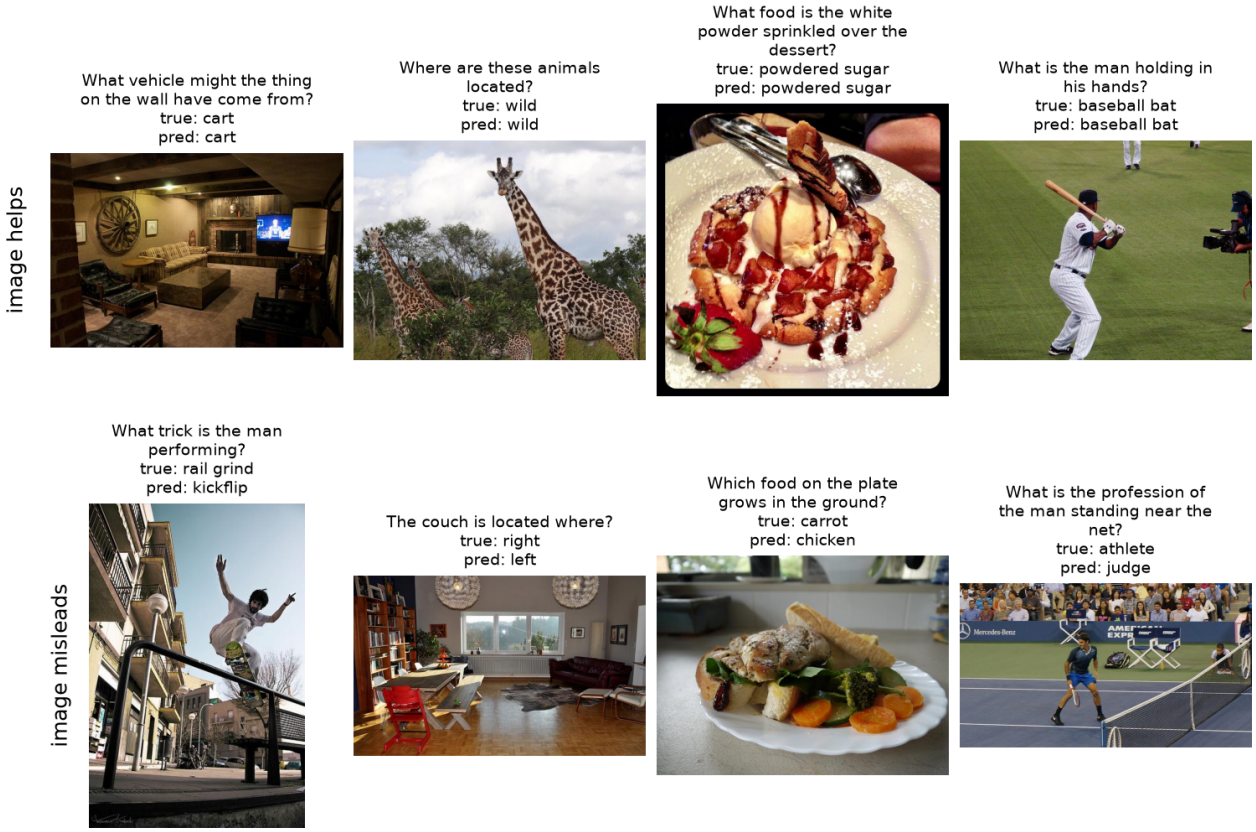


Figure 3: Seed 42. Top: the image helps (LateFusion right, TextOnly wrong; 263 questions). Bottom: the image misleads (TextOnly right, LateFusion wrong; 101 questions).

Support cases ask about something clearly visible; conflict cases contain a plausible wrong answer in the image (left vs right, two foods on one plate).

4 Status and next steps

- Done: data pipeline and split, feature cache, unimodal A, unimodal B and simple fusion with 5 seeds, shuffle test, support / conflict examples.
- For M3: the improved fusion (cross-attention from the choice tokens to the 50 image tokens), ablations (fusion level, product terms, fine-tuning the last CLIP block), error analysis by question type, compute cost, final report, slides and video.

5 AI Usage Disclosure

Claude Code (Claude Opus 5.5), used by Võ Thị Như Quỳnh on 9 November 2026, wrote the feature cache, the models and the training script, ran the experiments and drafted this report. Every number comes from saved run outputs read by `Assignment3/02_results.ipynb`; details are in `AI_USAGE.md`.

References

[1] D. Schwenk, A. Khandelwal, C. Clark, K. Marino and R. Mottaghi, "A-OKVQA: A benchmark for visual question answering using world knowledge," in *Proc. ECCV*, pp. 146–162, 2022. Dataset: https://huggingface.co/datasets/HuggingFaceM4/A-OKVQA, revision d1b0efa.

[2] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. ECCV*, pp. 740–755, 2014.

[3] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, PMLR 139, pp. 8748–8763, 2021.

[4] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems 30*, pp. 5998–6008, 2017.

[5] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.